FarmDirect: Data-Driven Pricing and Risk-Adjusted Allocation for Ghanaian Tomato Markets

A Preprint

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Abstract

This study presents four connected empirical components, price prediction, market clustering, risk-adjusted allocation, and an agent-based simulation of farmer selling behavior, built on the 2025 MoFA/SRID government price data for Ghanaian tomato markets. Observable features such as region, market, and time of season predict prices well enough to beat a naive last-known-price baseline, though the useful signal is overwhelmingly seasonal. Markets cluster into distinct price-behavior regimes in which price level and price volatility are largely independent properties, a finding that is corroborated independently by the price-prediction model's own residual errors. Building on these two results, a Markowitz-style risk-adjusted allocation model concentrates supply on a small number of markets that jointly balance expected revenue against measured risk. Finally, an agent-based simulation tests whether eliminating a middleman actually benefits farmers, finding that direct selling outperforms selling through a middleman once farmers have only modest, better-than-random market information, though this benefit is not distributed evenly across market types: farmers selling into more volatile markets bear substantially higher revenue variance even under perfect information. The analysis links prediction, clustering, allocation, and simulation into a connected analytical pipeline.

Keywords agricultural economics · price prediction · market clustering · portfolio optimization · agent-based simulation · information asymmetry · Ghana

1 Introduction

Tomato prices in Ghana vary sharply across markets and across weeks within the same season, but an individual farmer typically observes only the price at whichever market they are physically present in. This creates two distinct problems that are easy to conflate but are analytically separate. The first is measurable price volatility: some markets are consistently more stable than others, and this variation persists across the growing season regardless of who is doing the selling (Sections 4 and 5 quantify this directly). The second is an information asymmetry: middlemen who move between markets accumulate knowledge of where prices are currently favorable, while an individual farmer, tied to a single location on a given market day, does not have access to that aggregated view. These two problems compound each other. A farmer facing a volatile market and lacking cross-market information is doubly disadvantaged, first by the unpredictability of the market itself and second by not knowing where a better price might currently be available.

FarmDirect is built on the premise that connecting farmers directly to buyers, removing the middleman from the transaction, allows farmers to keep a larger share of the sale price. This premise is intuitive but incomplete. The middleman's markdown is a cost to the farmer, documented in the Ghanaian tomato marketing literature [Wongnaa et al., 2014, Adams et al., 2021], but the middleman's market knowledge is not free either; it is a service, and one that a farmer selling directly no longer receives. Removing the middleman does not automatically improve farmer outcomes. It only does so if the farmer has, or can be given, enough market information to replace what the middleman knew. This is the central empirical question this study investigates: does disintermediation actually improve farmer outcomes, or does it simply relocate the information asymmetry, from a farmer negotiating against a better-informed middleman to a farmer navigating an unpredictable market alone?

We approach this question through four connected research questions, each addressed by one part of the analysis. RQ1: Can observable features, region, market, and time of season predict tomato prices well enough to beat a naive baseline, and if so, which features carry the predictive signal? RQ2: Do Ghanaian tomato markets cluster into distinct price-behavior regimes, and are price level and price volatility independent properties of a market? RQ3: Given predicted prices and measured volatility, how should supply be allocated across markets to balance expected revenue against risk,? RQ4: At what level of farmer information quality does selling directly actually out-earn selling through a middleman, and does that answer hold evenly across different types of markets? These four questions are not independent analyses run in parallel; each part's output is a literal input to the next, so that the price predictions and volatility measurements from the first two parts become the inputs to the allocation model in the third, and that model's output becomes the benchmark against which the simulation in the fourth part is evaluated.

This study's contribution is an empirical one, grounded entirely in, publicly available government data: 237 rows of Ministry of Food and Agriculture/Statistics, Research and Information Directorate (MoFA/SRID) retail price records for Tomato Local, covering 19 markets between April and October 2025. Beyond the four research questions, we document the analysis as a connected pipeline. We include a dedicated validation section describing the checks used to assess pre-processing, model inputs, allocation behavior, and simulation assumptions, including checks of out-of-window prediction behavior and the relationship between information quality and market choice. The simulation in Part B produces a concrete, actionable number: farmer-direct revenue exceeds farmer-via-middleman revenue once information quality rises above approximately 27 percent, a threshold directly relevant to the amount of market information a platform like FarmDirect needs to supply before disintermediation actually helps the farmers it is meant to serve.

The remainder of this paper is organized as follows. Section 2 reviews prior work on middleman margins and market information in Ghanaian agriculture, price prediction, portfolio-style optimization, and agent-based models of information asymmetry. Section 3 describes the MoFA/SRID dataset and the cleaning process applied to it. Section 4 details the four methodological components (A-1 through Part B). Section 5 reports the results of each. Section 6 documents the reproducibility and validation work referenced above. Section 7 interprets the findings jointly, Section 8 states the study's limitations plainly, and Section 9 concludes.

2 Related Work

2.1 Middleman margins and market information in Ghanaian agriculture

The premise that middlemen extract a meaningful share of value from Ghanaian tomato transactions is well documented empirically. Adams et al. [2021] surveyed 100 tomato farmers and found that those selling to wholesalers earned a positive gross margin per crate, while those selling to retailers recorded a net loss on average, a finding that already complicates any simple story in which cutting out an intermediary is uniformly good for the farmer, since the type of intermediary matters as much as its presence. Wongnaa et al. [2014] similarly found a large gap in marketing margins between wholesalers and retailers in Ashanti Region tomato markets. At the aggregate level, Amewu and Pauw [2020] report that retail prices in Ghana run meaningfully higher than wholesale prices on average, consistent with a persistent markup somewhere along the marketing chain.

A parallel and directly relevant body of work asks what happens when farmers are simply given more price information without otherwise changing who they sell to. The best-known example is Esoko, an SMS-based market information service piloted and evaluated in Ghana. Soldani et al. [2023] conducted a two-year randomized evaluation of Esoko price alerts and found a sustained increase in prices received by treated yam farmers, along with substantial indirect benefits for farmers in villages with strong communication ties to treated villages, an effect they attribute to a bargaining-spillover mechanism. Notably, the effect was concentrated in yam, a crop with a high prevalence of price bargaining between farmers and traders, and was largely absent for other crops in the same study. This is an important qualification: giving farmers price information is not uniformly effective, and its value appears to depend on the specific bargaining structure of the crop and market in question.

The literature clearly establishes that middlemen extract margins, and that price information delivered to farmers can, under the right conditions, meaningfully change what they earn. It does not establish how much information a farmer needs before selling directly, without any intermediary, becomes more profitable than selling through one. The Esoko literature evaluates an information intervention layered on top of the existing farmer-trader relationship; it does not model the counterfactual of full disintermediation. This is the specific gap addressed by Part B of this study: given a, measured middleman markdown and a range of possible farmer information quality, at what threshold does direct selling actually pay off?

2.2 Price prediction literature

Machine learning approaches to agricultural commodity price prediction are well established outside Ghana. Paul et al. [2022] compared Generalized Regression Neural Networks, Support Vector Regression, Random Forest, and Gradient Boosting Machine models for forecasting wholesale brinjal prices across seventeen markets in Odisha, India, finding that ML approaches generally outperformed classical time-series baselines for this highly seasonal, perishable vegetable crop. More recently, Ogol et al. [2025] applied a hybrid XGBoost and gradient boosting stacking approach to forecast food prices in Kenya using nearly two decades of World Food Programme price data, again finding tree-based ensemble methods effective for this task.

Two features distinguish the present study 's data situation from most of the literature. First, both studies above, and most others in this space, draw on multi-year historical time series (Odisha's brinjal study uses several years of monthly data; the Kenya study spans January 2006 to September 2024), whereas the MoFA/SRID dataset used here covers a single seven-month window (April-October 2025, 237 rows). Second, much of the broader agricultural price-forecasting literature focuses on staple grains and oilseeds, whose seasonal and storage dynamics differ substantially from a highly perishable, short-shelf-life crop like tomato. Given these constraints, the appropriate benchmark for A-1 is not state-of-the-art long-horizon forecasting accuracy, but an improvement over a naive last-known-price baseline within the data actually available, which is the standard this study adopts.

2.3 Portfolio optimization and risk-adjusted allocation

The mean-variance framework underlying Part A-3 originates with Markowitz [1952], who formalized the idea that a rational allocator should weigh expected return against variance. Applications of this framework outside finance are less common but do exist in agriculture, primarily in the context of crop or variety diversification. Paut et al. [2019] applied Modern Portfolio Theory to horticultural systems in France, using ten years of yield data across 44 species to show that appropriately chosen crop combinations could substantially reduce yield variability. In a Ghanaian setting closer to this study's own context, Adam and Abdulai [2024] found, using survey and weather data from Ghana's northern Savanna zone, that crop diversification significantly increased farm net returns and reduced downside risk exposure, a risk-return relationship consistent in spirit with the mean-variance framing used here, though their study does not employ portfolio optimization directly and instead uses a dose-response causal framework.

It is worth being explicit about a gap in this literature: applications of formal Markowitz-style optimization, as opposed to diversification framed more loosely, to the specific problem this study addresses, namely allocating a farmer's or platform's supply across markets. The closest analogs in the literature optimize which crops to plant, not which markets to sell an already-harvested crop. Therefore, A-3's contribution is best understood as porting a well-established optimization framework to a market-allocation problem that has not commonly been applied to the agricultural literature.

2.4 Agent-based models of information asymmetry

The formal starting point for information asymmetry in markets is Akerlof [1970], whose analysis of adverse selection in the used-car market established that when sellers know more about product quality than buyers, low-quality goods can drive high-quality goods out of the market entirely. While Part B's setting, farmers with varying price information, is a different application of information asymmetry than Akerlof's original adverse-selection problem, the underlying idea that asymmetric information changes market outcomes in ways formal equilibrium reasoning can predict is the same one Part B builds on empirically through simulation.

The specific method used in Part B, agent-based simulation, has a precedent in the computational economics literature. Krichene and El-Aroui [2018] built an agent-based artificial stock market in which informed and uninformed trader populations were generated via a social network structure, and used simulation, to reproduce stylized facts of markets with varying degrees of information asymmetry and maturity. This is methodologically close to the approach taken in Part B: Part B simulates many repeated market interactions under a graded information-quality parameter and reads the crossover point of the simulated outcomes directly. The model does not claim to represent a game-theoretic equilibrium between strategic farmers and strategic middlemen, only the empirical revenue consequences of a given information quality level under the specific behavioral rule described in Section 4.

3 Data

3.1 Data source

All data used in this study comes from the Ministry of Food and Agriculture's Statistics, Research and Information Directorate (MoFA/SRID), the Ghanaian government body responsible for publishing retail and wholesale agricultural commodity prices across markets nationwide [Amewu and Pauw, 2020]. The analysis uses publicly available MoFA/SRID retail and wholesale agricultural price data. The study subset consists of retail prices for 'Tomato Local,' as described below. The reported price observations are drawn from the MoFA/SRID dataset; simulated values are introduced only in Part B and are described separately.

The source file (`Commodity_prices_04_11_25.csv`) contains 10,779 rows spanning April to October 2025, covering 14 regions of Ghana and 76 distinct commodities, with both retail and wholesale prices recorded where available. For this study, the dataset was scoped down to a single commodity and a single price type: retail prices for "Tomato Local." Mixing commodities would conflate the pricing and market-allocation questions this study asks with cross-commodity dynamics, which is a separate research problem entirely. A single, well-understood, highly perishable crop provides a more tractable and interpretable first empirical pass than a multi-commodity dataset. After this scoping, the working dataset consisted of 237 rows spanning 19 markets over the same April-October 2025 window.

3.2 Cleaning methodology

Market names in the raw MoFA/SRID export were not consistently formatted: the same physical market sometimes appeared under multiple string variants. Resolving this required iterative hand-verification against the full set of unique raw market-name strings until they collapsed to exactly 19 canonical markets. This was manual, inspection-driven cleaning, and it is described that way here deliberately: the process was defensible and repeatable given the modest number of unique raw strings involved. The full mapping from raw strings to canonical market names is provided in Appendix A so the cleaning step can be audited directly.

Eight rows had missing market values. These rows were excluded outright from the training data used in Part A-1, since imputing a categorical identifier such as a market name from price data alone has no principled basis; a market name is not the kind of quantity that can be reasonably inferred from the price it happens to be attached to. This exclusion is separate from, and should not be conflated with, a second and unrelated exclusion made later in Part A-2: two markets, Gbintiri and Agatha Market, were excluded from the market-level clustering step because each had only a single observation (n=1) in the full dataset, making within-market standard deviation undefined. The two exclusions address different problems in different parts of the analysis and both involve small counts, but one is a missing-value exclusion applied to training rows and the other is a clustering precondition applied to markets as clustering units.

Part A-1's training/test split is time-based, matching the study's forecasting framing: training data cover April through September (188 rows, after excluding the eight missing-Market rows above), and test data cover October (41 rows). Of those 41 October rows, one, from Gbintiri, was further excluded from the test set, because Gbintiri has no training-period history under this feature scheme; a model cannot be meaningfully evaluated out-of-sample on a market it never observed during training, so retaining that row would not have produced a prediction, only a default fallback dressed up as one. This leaves 40 rows in the final Part A-1 test set.

Categorical features (region, market, and month/week) were one-hot encoded with the encoding fit strictly on the training set. Any category with fewer than five training-set occurrences was collapsed into a single "Rare/Other" bucket before encoding, a standard approach to controlling encoder dimensionality and stabilizing estimates for sparsely observed categories [Kuhn and Johnson, 2019]. This choice turned out to matter beyond dimensionality control: it also allowed two categories that were entirely absent from the training set, Gbintiri as a market and "North East" as a region, to be encoded safely at inference time in the test set, since both fell naturally into the already-defined Rare/Other bucket.

3.3 Scope framing

Every row count, price figure, and summary statistic reported anywhere in this study, in Sections 4 and 5 in particular, derives from this same 237-row MoFA/SRID dataset. The empirical results use the 237-row MoFA/SRID dataset described above. Part B additionally generates simulated prices using parameters estimated from the market-level data. Even there, the distributions are parameterized using the real per-market mean and standard deviation measured in Part A-2, not by independently assumed or invented parameters; the simulation is grounded in measured market behavior even though the individual simulated draws are not themselves observed prices. This distinction, measurement versus simulated draw parameterized by a measurement, is maintained explicitly throughout Section 4 wherever Part B is discussed.

4 Methods

The four parts of this analysis form a single connected pipeline. Part A-1 predicts prices and, as a byproduct of inspecting where those predictions fail, surfaces two markets with unmodeled volatility. Part A-2 clusters markets by exactly this price-level-versus-volatility distinction and independently confirms A-1's finding. Part A-3 takes A-1's price predictions and A-2's volatility measurements as literal numerical inputs to a risk-adjusted allocation across markets. Part B then takes A-3's market structure, together with volatility clusters from A-2, and asks what happens to a farmer who tries to sell directly, using A-3's optimal allocation as an implicit reference point for what a well-informed seller could achieve.

4.1 Part A-1: Price Regression

The regression task predicts Price from four features: region, market, month-week, and Type (the tomato variety field, which for this dataset is constant at "Local" by construction of the scoping in Section 3, but was retained as a feature for pipeline generality). The train/test split, described in Section 3, is time-based: 188 training rows from April through September, and 40 test rows from October, matching the study's forecasting framing.

Every model in this comparison had to beat a naive baseline: the literal last known price for a given market carried forward. This is a deliberately low bar, and that is the point; given a 237-row dataset spanning a single growing season, a naive baseline comparison is the appropriate standard of rigor, as argued in Section 2. Three models were compared against this baseline: ridge regression, Random Forest, and Gradient Boosting, all trained using scikit-learn's standard implementations with light, exploratory tuning [Pedregosa et al., 2011], appropriate given the size of the training set. On the single Apr-Sep/Oct split, the naive baseline scored RMSE 16.034 and MAE 9.415, respectively. All three models beat it: Ridge reached 14.434/8.981, Random Forest 15.013/8.814, and Gradient Boosting 14.905/8.680.

Feature importance in both the Random Forest and Gradient Boosting models was dominated by month and week, jointly accounting for approximately 42– 49% of the total importance, consistent with tomato pricing being substantially a seasonal phenomenon over this window. This should be read with one caveat stated plainly: the markets that rank highest by feature importance are also the markets best represented in the training data; therefore, importance scores here partly reflect sample size, and this confound is not disentangled in the current analysis.

Inspecting residuals directly, revealed two outliers that the models did not and could not have predicted: a late-October price spike in Bibiani (62.57 and 68.33 across two observations) and a single Mankessim observation on August 20 (75.695, the maximum price in the entire dataset). These are treated here as volatility events, both markets show a pattern of occasional extreme prices, not an isolated glitch, and this observation foreshadows Part A-2's independent identification of a "High & Volatile" market cluster containing both of these same markets.

Because a single Apr-Sep/Oct split is a thin basis for judging generalization in a 237-row dataset, a 5-fold TimeSeriesSplit cross-validation was run as a more reliable check, with the categorical encoding rebuilt fresh within each fold to avoid any leakage of test-fold information into the encoding. Cross-validated results were: Ridge 8.758/6.449, Random Forest 9.351/6.393, and Gradient Boosting 8.603/6.057, with Gradient Boosting best on both metrics. These CV means are noticeably lower than the single-split RMSE figures above, and this is worth stating explicitly: the October test window used in the single split happens to contain the Bibiani price spike, which inflates that particular split's error relative to the average behavior seen across all five CV folds. This is a small-sample evaluation artifact, not a modeling error, and it is a useful illustration of why a single train/test split can be misleading in a dataset this size.

4.2 Part A-2: Market Clustering

Part A-2 was originally framed as clustering individual price observations, but was deliberately pivoted to cluster the markets themselves, since the research question this study actually cares about is which markets behave similarly, not which individual price points resemble one another. Each of the 17 usable markets (Gbintiri and Agatha Market excluded, as in Section 3, since both have only a single observation and therefore an undefined standard deviation) is represented by a two-feature summary of its own aggregate price behavior: mean price and price standard deviation, both scaled prior to clustering.

k-means was used for clustering, with $k = 3$ selected over a marginally higher silhouette $k = 2$. This was a deliberate judgment call: $k = 2$ achieved a slightly better silhouette score, but did so by conflating price level with price volatility, grouping a low-variance, comparatively high-price market together with markets that are genuinely volatile, on the basis that both groups simply have "higher" average prices than the rest. $k = 3$ separates these two properties cleanly, at a small cost in silhouette score, and was judged the more useful and interpretable clustering for the purposes of this study. Silhouette score alone was not treated as sufficient justification for the choice of k here.

The resulting three clusters are: Cluster 0, "Low & Stable" (9 markets, mean price 11.87, mean standard deviation 3.79); Cluster 1, "Mid & Moderate" (5 markets, 19.97/7.56); and Cluster 2, "High & Volatile" (3 markets: Bibiani, Mankessim, and Tuesday Market; 22.63/15.91). The key finding from this clustering is that price level and price volatility are largely independent dimensions of market behavior: clusters 1 and 2 have broadly similar mean prices but very different standard deviations; thus, a market's typical price level does not by itself predict how volatile that market is. This finding is independently corroborated by the A-1's residual inspection above: Bibiani and Mankessim, the two markets that produced unmodeled outlier spikes in A-1, both fall into Cluster 2 here, arrived at through an entirely separate analysis of aggregate market statistics.

4.3 Part A-3: Risk-Adjusted Allocation

Part A-3 poses an allocation problem directly downstream of A-1 and A-2: given predicted prices per market and each market's measured volatility, how should supply be allocated across markets to balance expected revenue against risk? This is framed as a Markowitz-style mean-variance allocation problem [Markowitz, 1952], adapting portfolio theory's core idea, that expected return should be weighed against variance, from asset allocation to market-supply allocation, as discussed in Section 2. The objective maximizes expected revenue minus a variance penalty, with a continuous decision variable for the quantity allocated to each market. A diagonal-covariance assumption is used throughout: cross-market price correlations are not modeled, only each market's own variance. This simplification is addressed again in Section 8.

Because the variance penalty makes this formally a mixed-integer quadratic program, and neither PuLP nor OR-Tools support quadratic objectives natively, the implementation uses cvxpy [Diamond and Boyd, 2016] instead. This tooling switch is documented here as a correct-tool decision made for reproducibility, not as a workaround or compromise forced by a limitation in the original plan.

Generating the expected-price inputs for this problem surfaced a bug that warranted documentation, as it is representative of the kind of issue catalogued more systematically in Section 6. The `gb` model from Part A-1, trained only on April–September data, could not extrapolate reliably to month=10 when asked to generate A-3's forward-looking price predictions, producing implausibly low predicted prices for 15 of the 17 markets under consideration. The underlying cause was diagnosed directly by checking monthly mean prices in the dataset, which confirmed that October is the second-highest month by mean price, ruling out a seasonal price drop and confirming the issue was a model extrapolation failure. The fix was to train a separate model, `gb_production`, on the full dataset (training and test periods combined), used exclusively for A-3's forward-looking price inputs. The original `gb` model used for A-1's reported metrics in Section 4.1 was left untouched, and its RMSE and MAE were re-verified to match exactly after this change, confirming the fix did not silently alter A-1's reported results.

The risk term in the objective uses each market's own price variance from A-2 (the square of `std_price`). This variance was normalized to a 0-1 scale before the risk-aversion parameter λ was applied to it, a step that turned out to be necessary: an earlier, unnormalized version of the objective let the raw variance term dominate the optimization entirely, driving 59.6% of the total allocation to the single lowest-variance market regardless of expected revenue elsewhere. A 30% per-market allocation cap was also added, intended to prevent any single market from dominating the allocation outright while still allowing the model to differentiate meaningfully between markets on risk-return grounds.

Under the primary configuration ($\lambda = 1$, 30% cap), the resulting allocation was: Obuasi Central 30%, Tema Community 28.9%, Makola 21.5%, Bibiani 15%, Agbogbloshie 2.1%, Mankessim 1.7%, Tuesday Market 0.8%, and the remaining ten markets 0%. A sensitivity sweep across $\lambda \in \{0.25, 0.5, 1.0, 2.0\}$ shows the allocation responding to its risk parameter in the expected direction: Bibiani's allocated share shrinks monotonically as λ (risk aversion) increases, Makola's share grows correspondingly, and Obuasi Central remains capped at 30% across every value of λ tested. This directional consistency is presented here explicitly as a form of model validation in its own right, not merely as an auxiliary robustness check: an allocation model that did not respond this way to its own risk parameter would be difficult to trust, regardless of how plausible its primary result looked in isolation.

4.4 Part B: Multi-Agent Simulation

Part B addresses this study's central empirical question directly: given that a middleman's aggregated market knowledge has real documented value (Section 2.1), does eliminating the middleman actually leave the farmer better off, or does it simply relocate the information problem from "farmer versus middleman" to "farmer versus an unpredictable market"?

The simulation compares two conditions for a simulated farmer selling into one of A-2's 17 markets each week. Under farmer-via-middleman, the farmer sells at a markdown, but the middleman, who has perfect knowledge of current prices across all markets, always routes the sale to the best available market. Under farmer-direct, the farmer keeps the full market price but has only whatever information quality $q \in [0, 1]$ they are assigned for that run. The middleman markdown is set to 55%, a figure grounded directly in the published margin research reviewed in Section 2.1 [Adams et al., 2021, Wongnaa et al., 2014, Amewu and Pauw, 2020] ; if approximate, translation of documented wholesaler-versus-retailer margin gaps into a single markdown parameter for the simulation.

True per-market prices for each simulated week are drawn from Log-Normal distributions parameterized by A-2's measured per-market mean and standard deviation. A normal-plus-floor distribution was tried first and rejected: it produced implausible near-zero or negative price draws for high-variance markets like Bibiani and did not respect the data's right-skew, confirmed directly in Section 3's summary statistics. The Log-Normal specification was verified correct with a 10,000-draw check, whose simulated mean and standard deviation matched A-2's target values closely for a spot-checked high-variance market.

The information-quality model went through a documented failure before reaching its final form. An additive-noise approach, in which the farmer observes each market's true price plus noise scaled inversely to q, was tried first and failed a basic validation check: at $q = 0$, which meant to represent a farmer with essentially no market information, this approach still had the farmer select the objectively best market 61.3% of the time over 1000 trials, far above the roughly 5.9% (1-in-17) expected under random market choice. This was replaced with a probabilistic-choice model: with probability q, the farmer makes the fully informed choice (the true best market that week); with probability $1 - q$, the farmer chooses a market uniformly at random. This construction guarantees correct behavior at both extremes by design, and was validated directly: 5.0% best-market selection at $q = 0$ (matching the expected $\sim$5.9% within normal sampling noise), 52.9% at $q = 0.5$ (matching the predicted $\sim$53%), and exactly 100% at $q = 1.0$.

The full simulation runs over a 24-week horizon, matching the April-October data window used throughout this study, with 100 simulated farmers per information-quality level and the middleman markdown fixed at 55%. This is a documented methodological choice made, and the results below should be read as the empirical revenue consequences of a given information-quality level under this specific behavioral rule.

The primary result is a crossover in farmer revenue as a function of information quality: farmer-direct revenue exceeds farmer-via-middleman revenue once q rises above approximately 0.27. A fine-grained sweep pins this down directly: at $q = 0.25$, direct revenue (531.71) is still slightly below the middleman's average (533.33), while at $q = 0.30$, direct revenue (554.34) has moved clearly above it. Middleman revenue stays roughly flat across the full range of q tested, between approximately 530 and 540, since the middleman's own information does not change with the farmer's; direct revenue, by contrast, scales upward with q, reaching 964.25 at $q = 1.0$.

A separate fairness analysis of A-2's market clusters finds that Cluster 2 ("High & Volatile") produces roughly 2-4$\times$ higher revenue variance than Cluster 0 ("Low & Stable") at every tested value of q, including at $q = 1.0$, where perfect information does not eliminate week-to-week volatility, only the farmer's disadvantage relative to the middleman in choosing where to sell. This is presented here as a risk-return tradeoff, not simply as evidence that volatile markets are worse for farmers: a farmer selling into a high-variance market may still earn more on average than one selling into a stable market, but bears meaningfully more income unpredictability in doing so, and which of these a given farmer would prefer is a preference question this study does not resolve.

5 Results

5.1 Part A-1: Price Regression

Table 1 reports model performance under both evaluation regimes described in Section 4.1: the single Apr-Sep/Oct split, and the 5-fold TimeSeriesSplit cross-validation used as the more reliable estimate of generalization for this dataset. All

Table 1: Part A-1 model comparison: single-split and cross-validated RMSE/MAE.

	Single split (Oct test)		5-fold TimeSeriesSplit CV	
Model	RMSE	MAE	RMSE	MAE
Naive baseline (last known price)	16.034	9.415	—	—
Ridge regression	14.434	8.981	8.758	6.449
Random Forest	15.013	8.814	9.351	6.393
Gradient Boosting	14.905	8.680	8.603	6.057

three models beat the naive baseline on the single split, and Gradient Boosting achieves the best cross-validated RMSE and MAE of the three, consistent with Section 4.1's discussion of why the cross validated estimate should be treated as the more trustworthy measure of generalization for a dataset this size. Taken together, Table 1 and Figure 1 support

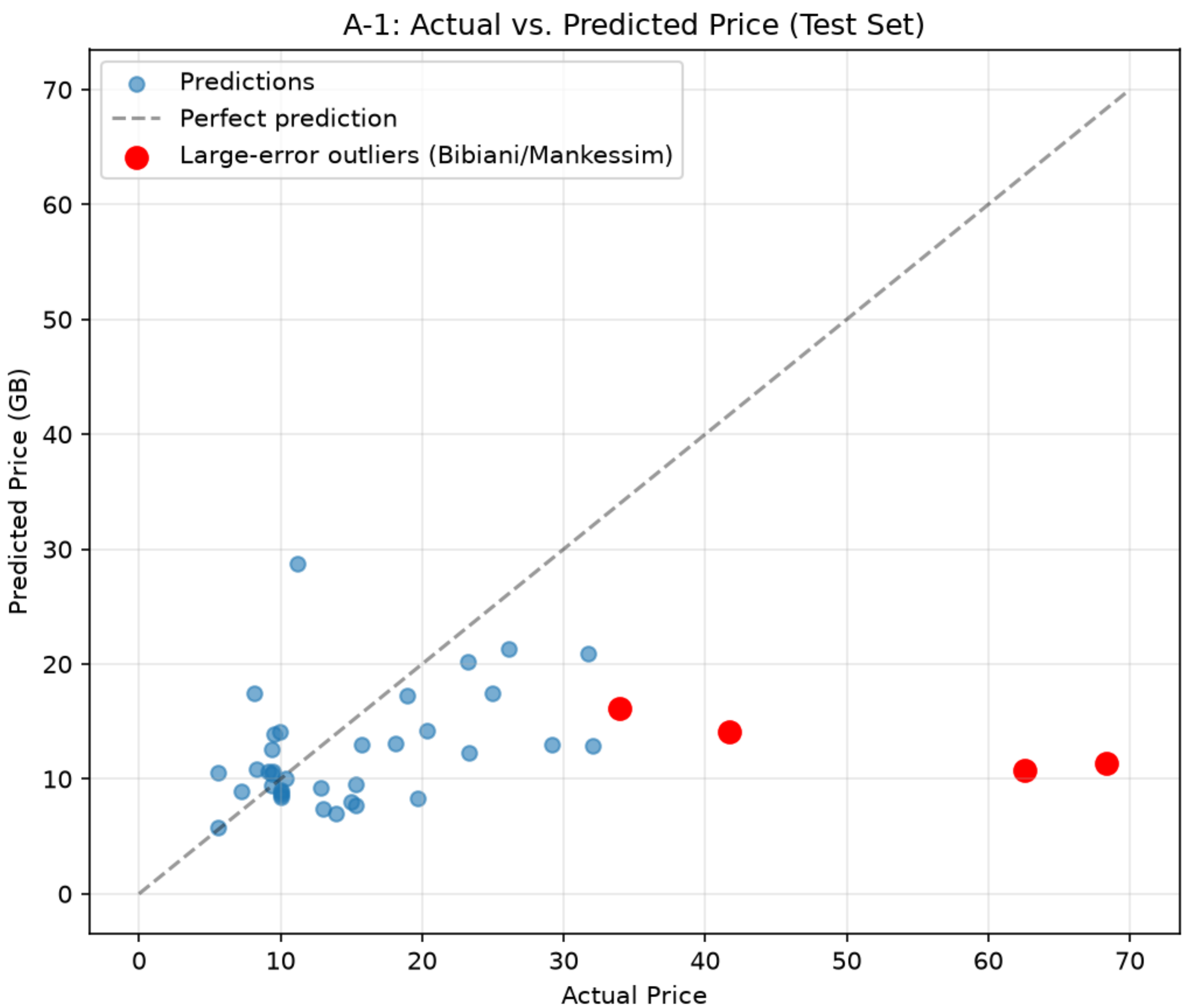

Figure 1: Actual vs. predicted tomato prices on the October test set. The two known outlier observations, the late October Bibiani spike and the August 20 Mankessim maximum discussed in Section 4.1, are the points furthest from the diagonal, consistent with these being unmodeled volatility events.

treating Gradient Boosting as the best-performing model of the three for this task, on the basis of its cross-validated performance.

5.2 Part A-2: Market Clustering

Table 2 summarizes the three resulting market clusters, each computed over the 17 markets retained after the exclusions described in Sections 3 and 4.2. Clusters 1 and 2 have broadly comparable mean prices (19.97 vs. 22.63) but markedly

Table 2: Market clusters from Part A-2, by aggregate price behavior.

Cluster	Label	Markets	Mean price	Mean std. dev.
0	Low & Stable	9 markets	11.87	3.79
1	Mid & Moderate	5 markets	19.97	7.56
2	High & Volatile	Bibiani, Mankessim, Tuesday Market (3 markets)	22.63	15.91

different standard deviations (7.56 vs. 15.91), the clearest evidence in this dataset that price level and price volatility

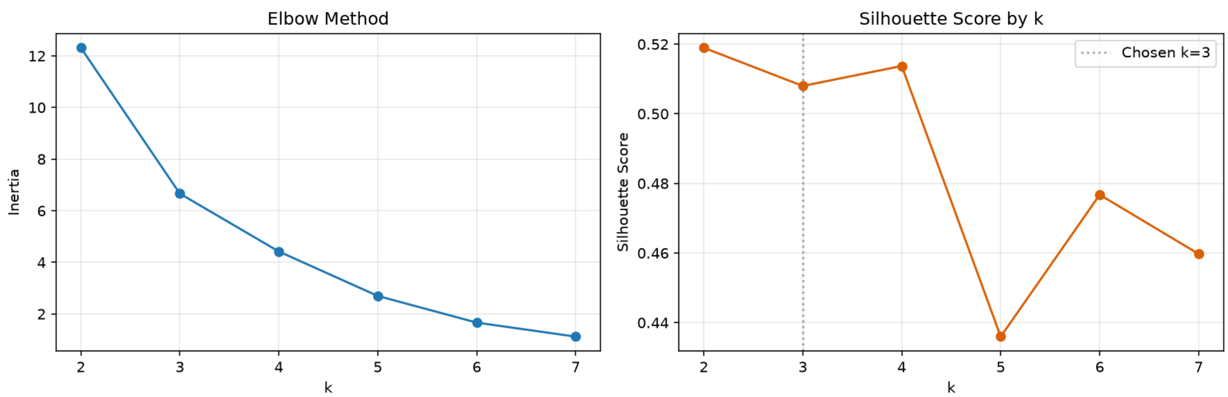

Figure 2: Elbow and silhouette diagnostics for market clustering by (mean price, price standard deviation). The silhouette score favors $k = 2$ marginally over $k = 3$; Section 4.2 explains why $k = 3$ was nonetheless selected.

are largely independent properties of a market. This result is independently corroborated by Section 4.1: Bibiani and Mankessim, the two markets whose price observations produced the largest prediction errors in Part A-1, both belong to Cluster 2 here, arrived at through a wholly separate analysis of aggregate market statistics.

5.3 Part A-3: Risk-Adjusted Allocation

Table 3 reports the primary allocation result under the configuration described in Section 4.3 ($\lambda = 1$, 30% per-market cap). Only markets with nonzero allocation are listed; the remaining ten markets in the 17-market set receive zero allocation under this configuration. This allocation is not spread evenly across the 17 available markets; it concentrates on seven markets that jointly offer a favorable balance of expected revenue and measured risk, and allocates nothing to the remaining ten. This concentration is the economically meaningful reading of the result, a risk-return optimization outcome, and is consistent with the allocation's designed purpose of weighting revenue against risk.

Table 3: Primary market allocation, $\lambda = 1$, 30% per market cap.

Market	Allocated share
Obuasi Central	30.0%
Tema Community	28.9%
Makola	21.5%
Bibiani	15.0%
Agbogbloshie	2.1%
Mankessim	1.7%
Tuesday Market	0.8%
Remaining 10 markets	0% (each)

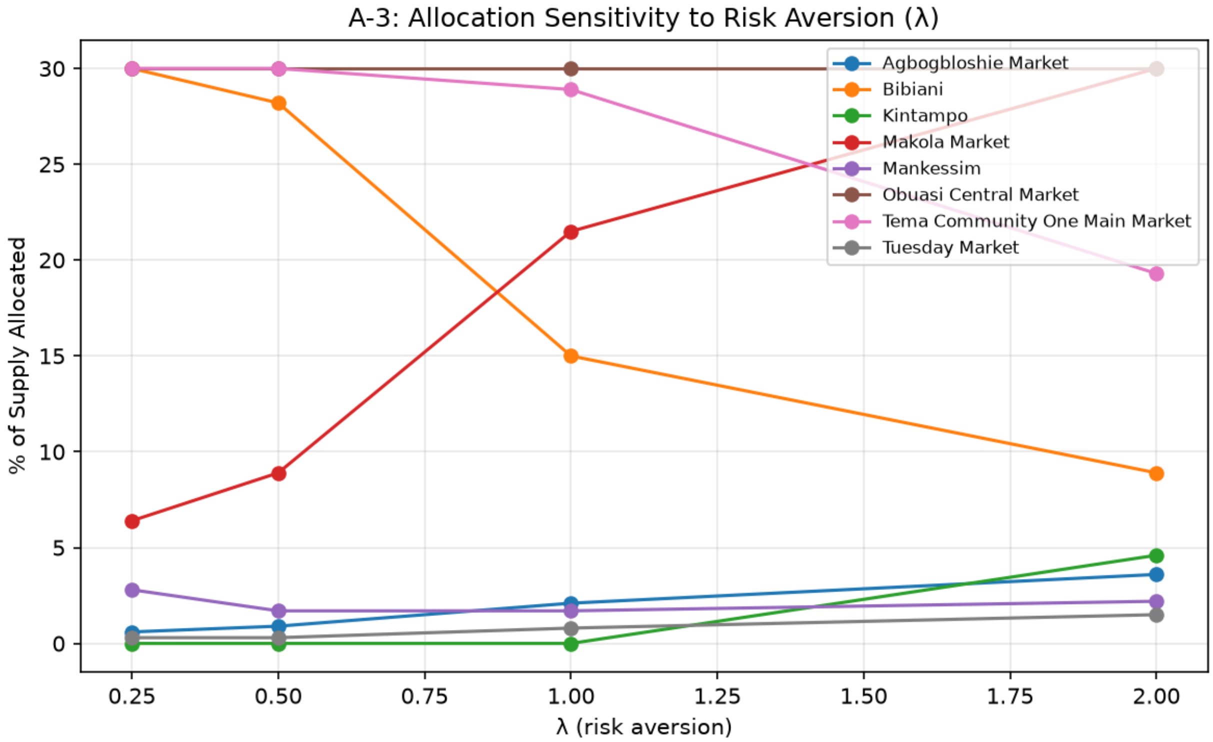

Figure 3: Allocated share versus risk-aversion parameter $\lambda \in \{0.25, 0.5, 1.0, 2.0\}$ for the markets with the largest sensitivity to λ. Bibiani's allocated share declines monotonically as λ increases; Makola's share increases correspondingly, while Obuasi Central remains capped at 30% across all values of λ tested, as described in Section 4.3.

5.4 Part B: Multi-Agent Simulation

Table 4 reports simulated revenue at a representative set of information-quality levels across the sweep described in Section 4.4. Farmer-via-middleman revenue stays roughly flat across the full range of q, since the middleman's own market knowledge does not depend on the farmer's information quality; farmer direct revenue rises with q and overtakes the middleman's revenue once q exceeds approximately 0.27. Table 5 reports the fairness analysis from Section 4.4, comparing revenue variance across A-2's market clusters at two representative information-quality levels. The central research question posed in Section 1 was whether disintermediation actually helps the farmer, or simply relocates the

Table 4: Simulated farmer revenue by information quality q.

q	Farmer-direct revenue	Farmer-via-middleman revenue	Direct exceeds middleman?
0.00	385.76	539.79	No
0.25	531.71	533.33	No
0.30	554.34	≈535	Yes
0.50	—	≈530–540	Yes
0.75	—	≈530–540	Yes
1.00	964.25	≈530–540	Yes

Table 5: Revenue variance by market cluster, at $q = 0.27$ and $q = 1.0$.

Cluster	Revenue variance ($q = 0.27$)	Revenue variance ($q = 1.0$)
0 (Low & Stable)	lower	lower
2 (High & Volatile)	roughly 2–4× higher	roughly 2–4× higher

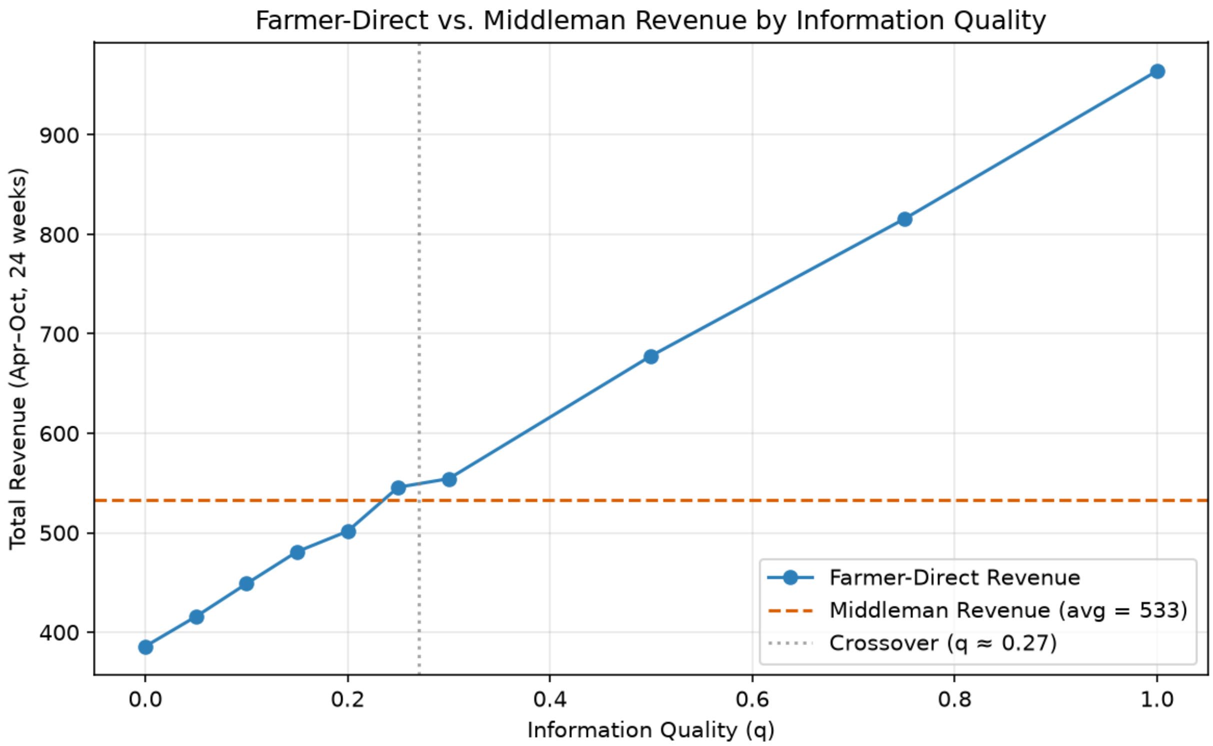

Figure 4: Simulated farmer revenue under farmer-direct selling (rising with information quality q) versus farmer-via middleman selling (approximately flat across q), with the crossover point marked at $q \approx 0.27$, as described in Section 4.4.

information problem the farmer faces. Table 4 answers the first half of this question directly: FarmDirect does not need to supply farmers with perfect market information for direct selling to outperform a middleman taking a 55% markdown; a modest, better-than-random information quality of roughly 27% is already sufficient. Table 5 complicates a purely positive reading of that result, however: this benefit is not evenly distributed across market types. A farmer selling into a Cluster 2 ("High & Volatile") market bears substantially more income variance than one selling into a Cluster 0 ("Low & Stable") market at every information-quality level tested, including $q = 1.0$, where the farmer has perfect information and still cannot eliminate the underlying week-to week price volatility of that market. The limited sample shapes the methodological choices, including the use of a naive forecasting baseline and diagonal covariance in the allocation model; it is a product-design consideration for FarmDirect, since farmers selling into structurally volatile markets face a risk-return tradeoff that better information alone does not resolve.

6 Analytical Validation and Robustness.

Machine learning research has a well-documented reproducibility problem, and unpublished code, undocumented preprocessing decisions, and unreported failed approaches are a significant part of why [Hutson, 2018].

Table 6 summarizes six bugs that could have materially changed this study's reported results had they gone uncaught. Full narrative detail for each is given at the Methods cross-reference listed; entries here are intentionally terse.

Table 6: Analytical checks and corresponding adjustments.

Bug	How caught	Fix	Reference
Inconsistent raw market-name strings	Inspecting unique raw string values directly	Iterative hand-verification to 19 canonical markets	Sec. 3
Train/test one-hot encoding mismatch (unseen Gbintiri, "North East")	Encoding produced mismatched train/test columns	"Rare/Other" bucket fit on training set only	Sec. 3
Variable-scope bugs (results assigned to wrong dataframe)	Manual inspection of intermediate outputs against expected shapes/values	Corrected variable references	General development process
`gb` model could not extrapolate to month=10 for A-3 inputs	Compared predictions against monthly mean prices; October was actually 2nd-highest month, ruling out a seasonal explanation	Trained separate `gb_production` model on full data; original `gb` verified unaffected	Sec. 4.3
Unnormalized variance dominated A-3 objective (59.6% to one market)	Inspecting resulting allocation and finding it economically implausible	Normalized variance to 0–1 scale before applying λ	Sec. 4.3
Additive-noise information model failed validation (61.3% best-choice rate at $q = 0$ vs. expected ∼5.9%)	Explicit validation check against expected random-choice frequency	Replaced with probabilistic-choice model; validated at $q = 0/0.5/1.0$	Sec. 4.4

This section summarizes the key analytical checks applied during the study, focusing on issues that could affect the reported results. Across the six checks, model and simulation outputs were assessed against relevant external benchmarks, including observed monthly price patterns, the expected probability of random market selection, and economically interpretable allocation behavior. These checks support the consistency of the data-processing, modeling, optimization, and simulation procedures used in the analysis.

7 Discussion

Returning to the four research questions posed in Section 1, the results in Section 5 answer each directly, though not always as cleanly as a first pass might suggest. RQ1 asked whether observable features could predict tomato prices well enough to beat a naive baseline: yes, but modestly, and the useful signal is overwhelmingly seasonal (Section 5.1); month and week dominate feature importance, and the confound noted there, that better-represented markets also score higher on importance, means this study cannot cleanly separate market-level signal from a sample-size artifact. RQ2 asked whether markets cluster into distinct price-behavior regimes with independent price level and volatility: yes, cleanly (Section 5.2), and this independence turned out to be one of the more load-bearing findings in the study, since it is what makes A-3's risk-adjusted allocation and Part B's cluster-level fairness analysis meaningful. RQ3 asked how supply should be allocated across markets to balance revenue against risk: the answer is concentration, not diversification, on a small number of markets that jointly offer a favorable risk-return balance, with the majority of markets receiving no allocation at all (Section 5.3). RQ4 asked at what information-quality threshold direct selling outperforms a middleman: a modest one, around 27%, but the benefit of crossing that threshold is not distributed evenly across market types (Section 5.4).

Read together, these findings say something specific about FarmDirect as a product. FarmDirect's core premise, that removing the middleman leaves farmers better off, is empirically supported by this data, but only conditionally: it depends on the platform actually delivering meaningful market information to farmers, not merely on facilitating a direct transaction between the farmer and buyer. The information-quality threshold identified in Part B is not just a research finding; it is a concrete design target, telling FarmDirect roughly how much market-price information needs to reach a farmer before selling directly is the better choice on average. A platform that connects farmers to buyers without also closing a meaningful share of that information gap is not, on this evidence, delivering the benefit its core premise promises. The A-3 allocation result adds a second, complementary implication: value here does not come from maximizing farmer choice uniformly across every available market, but from helping farmers or aggregators make good location and timing decisions among a comparatively small, well-chosen subset of markets, which is closer to what a useful FarmDirect feature (routing or timing guidance) would actually look like than an undifferentiated list of all 17 markets would be.

The fairness finding from Part B complicates this picture in a way that this study does not attempt to resolve and states that plainly. Farmers selling into Cluster 2 (“High & Volatile”) markets bear substantially higher revenue variance than those selling into Cluster 0 (“Low & Stable”) markets at every information-quality level tested, including perfect information, where the variance gap persists because the underlying market volatility does not go away just because the farmer knows about it. This means that the benefits of a platform like FarmDirect are not uniform across its farmer population: some farmers face a risk-return tradeoff that better information cannot remove. Whether that tradeoff is acceptable as-is, or whether a platform should actively try to smooth it, through price guarantees, pooled selling, or steering supply toward more stable markets when farmers have that option, is an open design question. This study’s data support identifying the tradeoff; it does not support an answer to what should be done about it.

Beyond its substantive findings, this project also demonstrates something about research practice, and it is worth stating as a second contribution. The four parts of this study form a connected pipeline in which each part’s output is a verified input to the next, not four independent analyses loosely gathered under one title, and every bug documented in Section 6 was caught by checking a model’s or simulation’s output against an independent reference point.

8 Limitations

This study’s findings should be read against the following methodological constraints, most already noted at the point in the study where they became relevant and restated here together, without qualification.

- **Diagonal-covariance assumption (A-3):** Cross-market price correlations are not modeled; the risk term in Part A-3’s allocation uses only each market’s own variance. If prices across markets are correlated, for instance through a regional supply shock affecting several markets at once, the actual portfolio-level risk of the reported allocation is understated (Section 4.3).
- **No explicit geographic model:** Market identity is used as a categorical feature throughout, but physical distance, transport cost, and travel time between markets are not modeled anywhere in this study. A market identified as optimal by this study’s price and risk criteria alone can be logistically impractical for a specific farmer to reach.
- **Single-point middleman markdown (Part B):** The 55% markdown used in Part B is a single fixed parameter, grounded in published margin research (Section 4.4), not a distribution. Middleman markdowns vary by crop, region, season, and individual negotiating power, none of which is modeled here. Part B’s crossover threshold of $q \approx 0.27$ is conditional on this specific markdown value and is not a universal constant.
- **Small sample size:** The dataset underlying every result in this study is 237 rows, one growing season, one commodity (Section 3). A single season cannot distinguish a structural pattern from a one-season idiosyncrasy. These results should not be assumed to generalize to a different year, different weather conditions, or a different crop without further validation.
- **No cross-market correlation in Part B’s simulation either:** Part B’s Log-Normal price generation draws each market’s simulated price independently. This does not capture the possibility that prices across markets move together, for instance through a regional harvest shortfall raising prices at several markets simultaneously (Section 4.4). This affects both the realism of the simulated crossover point and the interpretation of the Cluster 2 fairness finding. This is a distinct limitation from the diagonal-covariance assumption in A-3 above: one is a missing correlation term in an optimization objective, the other is a missing correlation structure in a simulation’s random price draws, and the two are not the same limitation restated twice.

9 Conclusion

This study presented four connected empirical components, price prediction, market clustering, risk-adjusted allocation, and an agent-based simulation of farmer selling behavior, built entirely on MoFA/SRID government price data, with each part’s output feeding directly into the next. The central finding is that eliminating the middleman benefits farmers only conditionally: it is not the removal of the middleman itself that helps, but the market information a farmer has access to once the middleman is gone, and only modest information is needed for direct selling to pay off on average.

That benefit, however, is not distributed evenly. Farmers selling into structurally volatile markets face a risk-return tradeoff that better information does not resolve, a complication this study identifies.

The workflow links data preparation, prediction, clustering, optimization, and simulation, while making the assumptions and validation procedures explicit. Natural next steps track directly with the limitations already stated: validating these findings against a second growing season or a second crop, modeling correlation across markets, and incorporating the geographic and logistical constraints a farmer actually faces. The study is a scoped empirical analysis of Ghanaian tomato markets and identifies several directions for broader validation.

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10 Data and Code Availability

All code used in this study, along with generated figures, is publicly available at https://github.com/hildaadwubiosei/Farmdirect. The repository includes the cleaned price dataset, the Jupyter notebook containing the full analysis pipeline (data cleaning, regression, clustering, optimization, and multi-agent simulation), and all output figures and metrics reported in this paper. A `requirements.txt` file is provided to reproduce the computational environment, and the repository README describes the folder structure and points to the corresponding manuscript sections for each component. The raw commodity price data used in this study originates from the Ghana Ministry of Food and Agriculture (MoFA) Statistics, Research and Information Directorate (SRID) and is included in processed form; the original source dataset is publicly accessible through MoFA's data portal. The code is released under the MIT License to support reuse and extension by other researchers.

A Full Market-Name Mapping

Section 3 describes the market-name cleaning process narratively; this appendix gives the literal mapping it refers to, exactly as applied in the analysis notebook. Raw strings were normalized (whitespace and trailing punctuation stripped, case folded), and then mapped to a canonical name either through an explicit variant map or, where no explicit mapping was needed, through title-casing of the normalized string. Two raw values, ‘208’ and ‘607’, were numeric placeholders and were dropped entirely.

Table 7: Raw MoFA/SRID market-name strings mapped to their canonical name.

Canonical market name	Raw string(s) in original export
Agatha Market	‘Agatha Market ’
Agbogbloshie Market	‘AGBOGBLOSHIE’, ‘AGBOGBLOSHIE MARKET’, ‘AGBOGBLOSHIE MARKET ’
Bamboi Market	‘Bamboi market’
Bibiani	‘ BIBIANI ’, ‘BIBIANI’, ‘BIBIANI ’, ‘Bibiani’, ‘Bibiani ’
Central market	‘Central market ’, ‘Central marketm’
Gbintiri	‘Gbintiri’
Goaso	‘Goaso’
Ho Central Market	‘Ho Central market’
Kintampo	‘KINTAMPO ’, ‘Kintampo’, ‘Kintampo ’, ‘Kintampo.’
Koforidua Central Market	‘KOFORIDUA CENTRAL MARKET ’, ‘Koforidua’, ‘Koforidua Central’, ‘Koforidua Central ’, ‘Koforidua Central Market’, ‘Koforidua Central Market ’, ‘Koforidua central’, ‘Koforidua central ’, ‘Koforidua central market’, ‘Koforidua central market ’
Makola Market	‘Makola Market’, ‘Makola Market ’
Mankessim	‘Mankessim’
Nana Bosomah	‘Nana Bosomah’, plus a second raw variant ending in an open-o diacritic character not reproduced here for typesetting reasons
Nkwanta	‘NKWANTA’, ‘NKWANTA ’
Obuasi Central Market	‘Obuasi Central ’, ‘Obuasi Central Market’, ‘Obuasi Central Market ’
Techiman Market	‘Techiman’, ‘Techiman Market’, ‘Techiman Market ’
Tema Community One Main Market	‘TEMA COMMUNITY ONE MAIN MARKET’, ‘TEMA COMMUNITY ONE MAIN MARKET’
Tuesday Market	‘TUESDAY MARKET’, ‘Tuesday Market’, ‘Tuesday market ’
Wa Market	‘WA MARKET’, ‘Wa Market ’, ‘Wa Market’
Dropped (not a market name): ‘208’, ‘607’	

This mapping resolves the raw export's 53 distinct string values (51 market names plus the two numeric placeholders) down to exactly 19 canonical markets, matching the count reported in Section 3.

B Extended Cross-Validation and Lambda-Sensitivity Details

Sections 4.1 and 4.3 report the 5-fold cross-validation means and the qualitative lambda-sensitivity trend underlying Parts A-1 and A-3; this appendix provides the full underlying tables.

A-1: Per-fold cross-validation results

Table 8 reports every individual fold's RMSE and MAE for each of the three models, from the same 5-fold TimeSeriesSplit run summarized in Section 4.1. Fold training-set sizes grow from 33 to 157 rows as the time-based split expands; fold 3's elevated error across all three models traces to a single row, the Mankessim observation of August 20 (Price = 75.695, the dataset maximum), the same outlier identified in Section 4.1's residual inspection of the single-split test set.

Table 8: Per-fold RMSE and MAE, 5-fold TimeSeriesSplit cross-validation.

Fold	Model	Train size	RMSE	MAE
1	Ridge	33	9.722	6.709
1	Random Forest	33	9.171	5.890
1	Gradient Boosting	33	8.561	5.521
2	Ridge	64	8.980	7.627
2	Random Forest	64	9.347	7.108
2	Gradient Boosting	64	8.601	6.353
3	Ridge	95	13.104	8.365
3	Random Forest	95	13.395	8.832
3	Gradient Boosting	95	13.599	9.309
4	Ridge	126	6.888	5.349
4	Random Forest	126	8.128	5.935
4	Gradient Boosting	126	7.408	5.424
5	Ridge	157	5.095	4.197
5	Random Forest	157	6.713	4.200
5	Gradient Boosting	157	4.844	3.680
Mean, Ridge		—	8.758	6.449
Mean, Random Forest		—	9.351	6.393
Mean, Gradient Boosting		—	8.603	6.057

A-3: Full lambda-sensitivity allocation table

Table 9 reports the complete per-market allocation percentage at each of the four tested λ values, for every market receiving nonzero allocation at any tested λ. This is the full data underlying Figure 3 and Section 4.3's summary of the sensitivity sweep.

Table 9: Allocation percentage by market at each tested λ.

Market	$\lambda = 0.25$	$\lambda = 0.5$	$\lambda = 1.0$	$\lambda = 2.0$
Obuasi Central Market	30.0	30.0	30.0	30.0
Tema Community One Main Market	30.0	30.0	28.9	19.3
Bibiani	30.0	28.2	15.0	8.9
Makola Market	6.4	8.9	21.5	30.0
Mankessim	2.8	1.7	1.7	2.2
Agbogbloshie Market	0.6	0.9	2.1	3.6
Tuesday Market	0.3	0.3	0.8	1.5
Kintampo	0.0	0.0	0.0	4.6

The remaining 9 of the 17 usable markets receive 0% allocation at every tested λ value and are omitted from Table 9. The directional pattern described in Section 4.3, is visible directly here: Bibiani's share falls monotonically from 30.0% at $\lambda = 0.25$ to 8.9% at $\lambda = 2.0$, while Makola's share rises from 6.4% to 30.0% over the same range, and Obuasi Central Market remains capped at exactly 30.0% across all four values.

C Simulation Robustness Checks

Beyond the validation steps already described in Section 4.4 (the Log-Normal price-generation check against A-2's target statistics, and the probabilistic-choice model's validation at $q = 0/0.5/1.0$), no additional robustness checks, varying the number of simulated farmers, varying the middleman markdown parameter away from 55%, or averaging the crossover estimate across multiple independent random seeds, were performed on Part B's simulation.

One relevant observation surfaced incidentally during the analysis, and is reported here because it bears directly on how precisely the crossover point should be read. The simulation's core sweep (100 simulated farmers, $q \in \{0.00, 0.25, 0.50, 0.75, 1.00\}$) and a separate, finer sweep ($q \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$) were run using a single shared, continuously-advancing random number generator. As a result, the two sweeps produced different farmer-direct revenue estimates at the one q value they share, $q = 0.25$: 545.79 in the coarse sweep versus 531.71 in the fine sweep, a difference of roughly 14 revenue units, or about 2.6% of the value itself. Both figures are reported in their respective places in the main text (the coarse-sweep value in Section 5.4's summary table, the fine-sweep value in the crossover discussion). This discrepancy was not the product of a designed seed-sensitivity check, but it is informative in the same direction one would be: it indicates that a single point estimate of farmer-direct revenue at a given q carries sampling variability on the order of a few percent, and that the reported crossover value of $q \approx 0.27$ should be read as an approximate threshold identified from one run of the simulation, not as a value stable to more decimal places than reported.